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# ECO344 - INTERNATIONAL ECONOMICS SUMMARY

Q. Briefly explain the term ―international trade‖.

International trade is the exchange of capital, goods, and services across international borders or territories because there is a need or want of goods or services. In most countries, such trade represents a significant share of gross domestic product (GDP). International trade allows countries to expand their markets for both goods and services that otherwise may not have been available domestically. As a result of international trade, the market is more competitive which results in more competitive pricing which brings a cheaper product home to the consumer.

Q. Discuss the theory of international trade.

International economics can be divided conveniently into two parts: real analysis or trade theory, and monetary analysis or international finance. Real analysis studies the reasons that trade takes place, the implications for commodity and factor price of changes in real variables (such as the stock of capital and the supply of labor), the benefits that accrue from international trade, and the effect of trade restrictions on the welfare of the economy. Because its focus is equilibrium determination of real trade flows and welfare, trade theory generally analyzes barter exchange expressed in terms of a numeraire good. It ignores macroeconomic disequilibrium problems by assuming the existence of full employment and aggregate trade balance. Monetary analysis, on the other hand, is concerned with such issues as the determination of exchange rates and the international transmission of unemployment and inflation. Often the two branches of international economics use different methodologies, with trade theory using market-clearing microeconomic equilibrium processes and international finance using macroeconomic concepts such as a single aggregate output and price level, in which there can be shortrun fluctuations. However, this distinction can easily he overdrawn. In recent years economists have made great strides in integrating the two approaches by modelling aspects of international finance, such as the existence of an aggregate trade deficit, as the result of microeconomic equilibrium processes in which agents trade goods both across borders and over time.

One question naturally arises: why is it necessary to distinguish trade between nations from trade between regions, and even from trade between individual consumers? The basic motivations for all such exchanges are similar, including differences in tastes and factor endowments. However, there are some unique features of international trade. First, though it is reasonable to assume that labor is completely mobile within a country, labor mobility among countries is severely restricted because of government regulation and differences in such things as language, religion, and social customs. Indeed, it is usually assumed in trade theory that labor is completely immobile among countries. Much of the theory of international trade also assumes capital to be immobile among countries, though we thoroughly analyze the implications of capital mobility later in the course of discussion in this course material. Differences in the degree of factor mobility are important because they help govern the incentives for and the implications of trade in commodities.

People in wealthier nations often argue that trade with poorer nations is harmful because it invites competition from low-wage foreign labor, while people in poorer countries make the opposite case that trade with countries with high-level technologies is unfair. These two views are fundamentally mercantilist in nature, in that they see international trade as taking place within a fixed-sum game. The gains to one country are accompanied by losses to another country. This view is wrong because International exchange, like trade among domestic agents, tends to expand aggregate incomes in all countries. Indeed, a substantial point of inquiry will be to investigate the nature of the gains from trade, or the benefits from international commerce.

Q. Briefly discuss the importance of international trade.

Globally, international trade has grown considerably in recent decades. For example, over the period between 1963 and 1979, the rate of expansion of real merchandise exports (that is, the value of exports deflated by changes in export prices) in the world averaged 11.8 percent per year, a remarkably high growth rate by historical standards. Indeed, this figure likely underestimates the true growth in the real volume of exports because available price data do not adequately account for the marked improvements in product quality in recent years. At the same time, global growth in real output, measured by gross domestic product (GDP) in each country, averaged 6.1 percent per year, also high by historical standards. Thus, during that period, the world experienced a rapidly rising effective integration among countries as they become more closely interrelated through international trade in goods. This trend continued after 1979,

though economic activity grew at markedly slower rates. Over the period between 1979 and 1991, real export growth averaged 4.4 percent per year, while real output expansion averaged 2.9 percent per year.

Consider the data in the final two columns of Table 1.1, which show the ratios of merchandise exports to GDP in 1970 and 1991. These ratios are often considered to be measures of a nation's —openness|| to international trade, though it is more appropriate to interpret them straightforwardly as indications of the share of national production that is exported. Thus, they provide rough suggestions of the relative importance of international trade in aggregate output. In Uganda, as in some other very poor nations, this export share has fallen considerably over the last 25 years, because of a dramatic decline in Uganda's merchandise exports. On the other hand, with the exception of Japan, the East Asian economies in our table registered marked increases in the contribution of their exports to GDP. Most striking is the experience of China, whose exports rose explosively from 1.8 percent of GDP to 19.5 percent of GDP. That .Japan's share was relatively static does not mean that export growth was unimportant. To the contrary, Japan's merchandise exports rose sixteen-fold over the period, as did its GDP. No other developed nation experienced such rapid increases in economic activity. Thus, at this level it appears that rapid trade growth is positively related to rapid economic growth.

Q. Discuss the differences between import trade and export trade.

Import means to buy goods and services from a different country to the home country. Therefore, these goods and services are those which are produced in a foreign land and are bought by the particular domestic country. They can be shipped, sent by email, or even hand-carried in personal luggage on a plane.

When a particular country does not have a particular good or resource, it is necessary to import that good or resource from another country. Most countries import raw materials or commodities that are not available within its borders. The best example of this is how a lot of countries import oil from the Middle Eastern countries. Therefore, the beneficial party in imports is the party that exports those products.

Export, on the other hand, is sending goods or services to another country for sale. As a result, it brings in the foreign income to the domestic country. The sale of such goods adds to the producing nation's gross output. For example, when a country manufactures excess of some products or if it has

more natural resources like oil, that country can sell them to other countries and obtain an extra income.

The main difference between import and export is that the import refers to bringing goods and services from other countries to the home country while the export refers to selling goods and services from the home country to other countries.
Export and import are essential phenomena in the international economy. Both these trading processes directly affect the economy, facilitating the economic advancement in a country and the world as a whole.

Q. Discuss the sectorial structure of trade.

**The Sectorial Structure of Trade**

Levels of trade can be significant in particular sectors of the economy even if the overall trade ratios are modest. For instance, the United States imports all of its consumption of certain tropical products, such as cocoa. Looking at two major domestic sectors, in 1990 the United States exported over 45 percent of its agricultural production and imported over 43 percent of its consumption of motor vehicles and automobile parts. Clearly, changes in the international economy that affect these sectors bear potentially significant impacts on domestic prices, output, and employment. Further, such impacts can spill over into other portions of the economy through their effects on consumer demand and input purchases.

Some brief comments about the sectors in this table are in order. It is evident that export strength in food is related to the existence of abundant supplies of agricultural land, as is found in China, Canada, and the United States. Correspondingly, countries with limited land supplies, such us Korea, Japan, and Switzerland, tend to import food. Similar statements can be made about the determination of exporters and importers of fuel. The United States is noteworthy in that although it is one of the largest petroleum producers in the world, it remains a major importer because of its huge demand for energy. In general, however, it is clear that relatively greater supplies of natural resources area major determinant of comparative advantage.

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The remaining three sectors-chemical, automotive products, and office machines and telecommunications equipment-all represent relatively sophisticated manufacturing products. In addition to standard inputs in production, these goods tend to require substantial scale, innovation, and product differentiation for export success. The developed countries compete among themselves in the latter dimensions, so that there is no obvious pattern of comparative advantage for these goods within that group. For example, Germany, the United Kingdom, the United States, and Switzerland are all major exporters of chemicals, while Canada and Japan tend to Import them, as do the developing countries. Germany, Japan, and Spain are successful exporters of automotive products, while the United States Mini Switzerland are major net importers. Comparative advantage in office machines and telecommunications equipment is similarly mixed, with Korea and Singapore having broken into the ranks of net exporting countries.

In principle, trade in services should be treated no differently from trade in goods. Some countries, depending on their factor supplies, technology, and tastes, have a comparative advantage in providing certain services to international customers, just as some countries have a comparative advantage in certain goods. Major traded services include financial or management expertise, insurance underwriting, transport, tourism, construction, and numerous other professional services. Nonetheless, some important distinctions between trade in goods and services arise. For example, sometimes foreign purchasers come to the domestic economy to consume a service, such as a medical procedure or a vacation. These transactions are properly regarded as exports for the providing country. On the other hand, to provide banking services in a foreign market typically require establishing facilities there instead of exporting some tangible commodity. Because the banking services are produced in the foreign market using primarily foreign inputs we would not count them as exports for the country undertaking the investment.

Q. Discuss the analysis of Global Economy in the world

In recent years international economic issues have taken center stage in the news. For example, on January 1, 1994 the United States, Canada, and Mexico entered into a joint compact, called the North American Free Trade

Area (NAFTA) that would gradually reduce trade barriers among them. As readers may recall, negotiation of NAFTA was heavily controversial in all three nations, and its passage was anything but certain. Some people in the United States were worried about the impact of freer trade with Mexico on the living standards of lower-skilled Americans, while others had concerns about the potential effects of NAFTA on environmental standards in the region. On the other hand, U.S. advocates of the agreement proclaimed its potential to raise incomes overall through greater trade and investment flows. Canadians had the same concerns and hopes about the potential effects of NAFTA, with further worries about safeguarding the security of their supplies of oil and natural gas. For their part, many Mexicans were wary of closer competition with the United States' high productivity standards and advanced technologies, expressing particular concern about the fate of traditional Mexican agriculture and peasant cultures.

The countries of the world are also moving toward closer trade integration through acceptance of the Uruguay Round Agreement in the General Agreement on Tariffs and Trade (GATT). The GATT agreement would set out broad rules governing national policies that influence international competition, including tariffs, quotas, foreign investment regulations, agricultural subsidies, and patents and copyrights, among other practices. Because different countries have conflicting interests in these areas, negotiation of the Uruguay Round accords was also quite contentious. Nonetheless, most economists argue that its passage will represent a valuable step forward for global trading relations, bearing the potential for expanding trade and world incomes by hundreds of billions of dollars per year.

As nations have become more interdependent in recent decades through growth in international trade and investment, episodes of trade conflict have become more evident and interesting to the public. An obvious example is the continuous effort by the United States and Japan to manage their bilateral trade relationship, which involves a significant American trade deficit with Japan. Many American critics claim that the Japanese market is not effectively to foreign firms, while a standard Japanese response is that foreign firms do not try hard enough to penetrate the market. Japan is hardly unique in this regard, of course. There are loud complaints from numerous countries about protectionism and arbitrary government interference with trade in the United States, Canada, South Korea, India, China, and the European Community, among other nations and areas. Unquestionably, there are significant pressures in the world economy for nations to interfere with the free exchange of goods in order to limit the negative effects trade may have on some groups and industries.

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A major component of this growing international interdependence is the phenomenal growth of multinational enterprises (MNEs), firms that have production and marketing facilities in numerous countries. Many global corporations have become absolutely huge in terms of world sales, assets, and employment, and their international operations have significant effect on both host and home nations. Accordingly, these firms are highly controversial in a number of dimensions, with some people blaming them for shifting jobs out of high-wage countries to low-wage countries and other claiming that they change locations in response to differences in environmental or business regulations. On the other hand, most economists tend to view MNEs as conduits for efficient global allocation of capital.

Q. Describe the term _gains from trade‘

In economics, gains from trade refers to the net benefits to agents from allowing an increase in voluntary trading with each other. In technical terms, it is the increase of consumer surplus plus producer surplus from lower tariffs or otherwise liberalizing trade. Gains from trade are commonly described as resulting from specialization in production from division of labor, economies of scale, scope, and agglomeration and relative availability of factor resources in types of output by farms, businesses, location and economies, a resulting increase in total output possibilities, trade through markets from sale of one type of output for other, more highly valued goods.

Q. List the two divisions of gains from trade

Q. List and explain the factors affecting gains from trade.

**Factors Affecting Gains from Trade**

1. Differences in cost ratio: The gains from international trade depends upon the cost ratios of differences in comparative cost ratios in the two trading countries. The smaller the difference between exchange rate and cost of production the smaller the gains from trade and vice versa.

2. Demand and supply: If a country has elastic demand and supply gains the gains from trade are higher than if demand and supply are inelastic.

3. Factor availability: International trade is based on the specialization and a country specializes depending upon the availability of factors of production. It will increase the domestic cost ratios and thereby the gains from trade.

4. Size of country: If a country is small in size it is relatively easy for them to specialize in the production of one commodity and export the surplus production to a large country and can get more gains from international

trade. Whereas if a country is large in size then they have to specialize in more than one good because the excess production of only one commodity cannot be exported fully to a small sized country as the demand for good will reduce very frequently. So the smaller the size of the country, the larger the gain from trade.

5. Terms of Trade: Gains from trade will depend upon the terms of trade. If the cost ratio and terms of trade are closer to each other more will be the gains from trade of the participating countries.

6. Productive Efficiency: An increase in the productive efficiency of a country also determines its gains from trade as it lowers the cost of production and price of the goods. As a result, the country importing gains by importing cheap goods.

Q. Discuss the relationship between the supply and production function

An understanding of trade requires an understanding of complex and indirect relationships, such as how a country's endowments of capital and labor determine its optimal pattern of trade. Before we can grasp the whole picture, we need to establish an understanding of the individual pieces of the puzzle. In this unit, we will develop the tools of production theory and producer equilibrium. Many of you will be familiar with the basic ideas from intermediate microeconomics or principles of economics. For those of you who are relatively unfamiliar with these technical constructions, we urge you to work through them slowly and carefully.

The production relation described by Eq. (2.1) contains three variables: the levels of input of capital and labor and the level of output. Geometrically, it could be represented by a three-dimensional surface; diagrammatically, it can be illustrated as in Fig. 2.1. The production surface can be thought of as it a hill, with the origin representing the ground level. At the origin there is neither labor input nor capital input, and therefore, there is no output. With positive amounts of both capital and labor, there will be a positive level of output of X, and as we add more of either capital or labor or both, the output of X increases.

Q. Explain what you understand by market incentives

Market incentives, such as reflected in prices of outputs and inputs, are theorized to attract factors of production, including labor, into activities according to comparative advantage, that is, for which they each have a low opportunity cost. The factor owners then use their increased income from such specialization to buy more-valued goods of which they would otherwise be high-cost producers, hence their gains from trade. The concept may be

applied to an entire economy for the alternatives of autarky (no trade) or trade. A measure of total gains from trade is the sum of consumer surplus and producer profits or, more roughly, the increased output from specialization in production with resulting trade. Gains from trade may also refer to net benefits to a country from lowering barriers to trade such as tariffs on imports.

Q. Discuss the analysis of return to scale

A very common assumption in economics is that of constant returns to scale, the assumption that proportional changes in all inputs lead to the same proportional change in output. This assumption is referred to somewhat more formally as homogeneity of the first degree. It is such an important concept in economics and in the discussion of this material that a formal definition seems worthwhile.

Definition. Let $\lambda > 0$. The function $X = F(K, L)$ is said to be homogeneous of degree k if $\lambda X = F(\lambda K, \lambda L)$. If $k = 1$, the function is said to be homogeneous of degree 1, and production is characterized by constant returns to scale.

The returns-to-scale assumption can be illustrated by the isoquant diagram of Fig. 2.4. Consider first the isoquant X0, where it has been assumed that the level of output is equal to 10. All the points on the curve X0 represent the infinite number of combinations of capital and labor that, when combined with the assumed technology, would give this level of output. One such point, A, shows that 10 units of output can be produced using 6 units of labor and 7 units of capital. Suppose we now double the inputs of both factors and move to point B. Since the capital/labor space is —full‖ of isoquants, there must be one that passes through point B. The question now is: What is the level of output associated with that particular isoquant? Clearly, the answer depends on the assumption we make about the degree of homogeneity of the production function. If k is equal to 1, yielding constant returns to scale, then it is clear that the level of output associated with point B must be 20 units, twice that associated with point A. If, on the other hand, k is greater than 1, implying increasing returns to scale, then although we do not know the precise number to be attached to isoquant X1, we know that it will be greater than 20. Similarly, for decreasing returns to scale, where k is less than 1, the level of output associated with X1 would be less than 20. Another characteristic associated with the concept of homogeneity relates to the slopes of the isoquants as we move along a ray from the origin, as in K/L of Fig. 2.4. It can be shown that for any such ray, and for

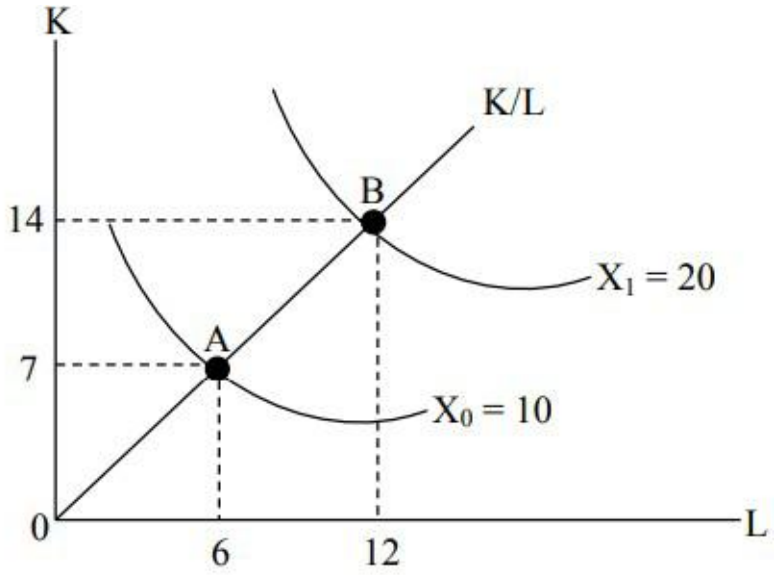



Fig. 2.4: Constant Returns to scale

Q. Explain the effects of international trade on underdeveloped countries.

**Consequences/Effect Of International Trade**

1. When there is Dual Economies

International trade has resulted in creating ‗dual economies' in underdeveloped countries as a result of which the export sector became an island of development while the rest of the economy remained backward. The effects of foreign factor movements have been that of creating a highly unbalanced structure of production of these countries. No doubt, the opening up of the export markets gave a fillip to their export sector which led to the development of this sector while ignoring other sectors of the economy.

2. When the Poor Countries Benefit Nothing

The foreign trade has also not been entirely beneficial to poor countries because of the adverse effects of foreign investments on their economy. It has been maintained that the inflow of foreign capital and developed a country's natural resources only for export purposes, to the neglect of production in the domestic sector. In these countries the export sector remains an island of development surrounded by a backward low-productivity sector. Thus, the inflow of foreign capital in underdeveloped countries has not resulted either in the development of the domestic sector

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or of the people in these countries. Despite huge foreign investments, the people have remained backward in their countries.

3. When there is Limited Possibility of Gain

According to Prof. Nurkse the possibility of gain from foreign trade to underdeveloped countries is restricted or limited. It is simply due to the reason that underdeveloped countries export mainly primary goods. These exports suffer losses on account of:

(i) Fall in the demand due to the tendency on the part of developed countries to establish heavy industries

(ii) Contribution of services in the aggregate production of developed countries has been increasing

(iii) Income elasticity of demand for agricultural production is less in developed countries

(iv) Many developed countries have been adopting policy of protection in respect of agricultural products

(v) Use of synthetic goods in place of agricultural products has been on the increase.

On account of these reasons, income of underdeveloped countries from the export of primary products has been diminishing constantly. Under these circumstances, it is totally wrong to call trade as _an Engine of Growth'.

4. When there is an Adverse Effect on _Demonstration Effect'

Another harmful effect is that the international operation of the _demonstration effect' has been a handicap for the poor countries. It has been responsible for reducing the capacity for capital formation. The desire for luxury, show-off for higher standard of living and patterns of consumption of advanced countries has been an important factor responsible for low level of domestic savings in underdeveloped countries. Higher income groups in these countries are trying to adopt the consumption standards of advanced countries which have pushed up their propensity to consume and thereby limited capital accumulation and economic growth. This leads to corruption and black marketing. Thus, these evils have adverse effect on the economy.

5. When there is Secular Deterioration in the Terms of Trade

In the opinion of Prof. Raul Prebisch, there has been a secular deterioration in the terms of trade of underdeveloped countries. How maintains that

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underdeveloped countries have suffered with fatal effects of a continuous weakening in their capacity to import. It has lead to the weakening of the capacity of their existing primary producing industries to support their growing population. It has resulted in a failure to transmit to them the benefits of technical progress.

Q. Write short notes on the following concepts

(i) Productivity of labour

It is perhaps most natural to consider, capital to be fixed in its sectorial usage for some time, and this is the basic approach we adopt. However, we note that labor could also be characterized similarly to the extent that skills stock of Rr , but it does so at a diminishing rate in if LXI, -u some point, the total product, could fail?. In the bottorr panel wo depict, the marginal product of labor in industry X, which is the slope of the total product curve. Because of diminishing returns, the marginal product declines as employment in X rises. A similar construction applies to good Y, given the fixed stuck of Sy.

These results hold in a more general sense. The marginal product of labor in either sector is an increasing function of the capital – labor ratio used in that sector. Consider, for example, an increase in usages of 10 percent and an increase in labor usage of 5 percent in industry X. We may decompose this change into, first, an identical 5 percent rise in R and L and, second, an additional rise in R. With constant returns to scale, the first component has no impact on the marginal product of labor. However, the second component raises the marginal product of labor for any given level of employment. Indeed, as depicted in Fig. 9.1, an increase in the endorsement of Rx will shift up both the total product and marginal product curves for labor.

(ii) Factor Price Equalization theorem

The Stolper–Samuelson theorem is closely linked to the factor price equalization theorem, which states that, regardless of international factor mobility, factor prices will tend to equalize across countries that do not differ in technology.

It is important to examine this result further. The Stolper-Samuelson model holds factor endowments fixed while allowing commodity prices to change, which shifts the demands for factors, thereby changing real factor incomes. This is not overly restrictive, for it seems reasonable to suppose that factor supplies arc exogenously given. However, as we suggested earlier, there may be some question about this if, for example, labourers are able to clause between work and leisure.

(iii) Taxes and subsidies

The examination of taxes and subsidies is intended as an example of the effects that a wide range of government policies can have on trade. For example, environmental policies and regulations impact on firms' costs, therefore having effects on outputs and trade similar to those produced by taxes. Government policies such as taxes and subsidies will generate trade. However, they show equally clearly that, trade induced by the introduction of distortions is not beneficial trade.

Q. Discuss the following:

(i) Trade mid factor prices

In the specific-factors model, the equalization of commodity prices by international trade; does not equalize factor prices, An important implication of ibis finding is that free trade in goods does not fully exhaust the available gains from trade, in the sense that, productive factors have an incentive to migrate in such an equilibrium. Even in the absence of any impediments to trade, if factors are intersect orally immobile, there will be international differences in their real returns. Factors may be too impatient to wait for the long-run equalization of their returns (assuming such equalization were even possible in practical terms) and may prefer to find employment in their sections.

(ii) Endowment Changes and Outputs

Turning to the Rybezynski theorem, our earlier discuss on it in the previous units demonstrated that under constant commodity prices, an increase in the endowment of a particular factor raises the output of the good intensive in that factor and lowers the output of the other good. In the specific-factors model, where factor intensity is less important than factor specificity, the predictions about endowment changes and outputs are again quite different, as we have already demonstrated.

Specifically, we have shown that an increase in the endowment of a specific factor increases the output of the good that uses that factor and must lower the output of the other good by pulling labor from it. This is similar to the Rybczynski prediction, but it depends on the sectorial specificity of the expanding factor rather than on factor intensities in-production. We have also demonstrated that a rise in the labor endowment expands both outputs as the new-labor force is divided between the sectors. Again, this is due to the mobility of labor rather than to the intensity of production. Thus, we can state the following proposition.

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(iii) Factor endowments and outputs

An increase in one specific factor increases the output of the commodity that uses that factor and reduces the output of the other industry. Increases in the supply of the mobile factor will expand both outputs.

We note in passing that the impacts of endowment changes in the specific-factors model suggest that different groups of factors will have different incentives to lobby for or against international factor migration. It is clear from our discussion that mobile factors would oppose policies allowing freer immigration of competing mobile factors, while owners of specific factors would favor them. Again, this is a result of factor mobility rather than factor scarcity, as in the Heckscher-Ohlin model.

(iv) The Pattern of Trade in a specific-factor model

The pattern of trade in the specific-factor model, each country will export the good with the absolutely abundant stock of specific capital, assuming identical endowments of labor, the mobile factor. With differences in labor endowments, trade patterns will depend on the nature of the production functions and on the allocation of capital (that is, on the stocks of specific factors.

(v) Factor intensity

In the specific-factors model, where factor intensity is less important than factor specificity, the predictions about endowment changes and outputs are again quite different. It is also possible to allow moderate differences in demand patterns between countries, provided these differences are not so great as to overcome the effects of endowments. Finally, it is theoretically possible for a factor-intensity reversal to occur between countries. That is, given the differences in relative factor prices between nations, product Y could be capital-intensive in capital-abundant Country. H and yet be labor-intensive in labor-abundant Country F in autarky. In ' this case, the Heckscher-Ohlin theorem would predict that both countries would export the same good, a practical impossibility.

Q. Differentiate between Rybezynski theorem and Factor content theorem

The Rybczynski theorem was developed in 1955 by the Polish-born English economist Tadeusz Rybczynski (1923–1998). It states that at constant relative goods prices, a rise in the endowment of one factor will lead to a more than proportional expansion of the output in the sector which uses that factor intensively, and an absolute decline of the output of the other good. The Rybczynski theorem explains the outcome from an increase in one

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of these factor's supply as well as the effect on the output of a good which depends on an opposing factor. The Rybezynski theorem describes the second fundamental relationship in the economy. Named for the British economist who developed it, the theorem is concerned with the relationship between changes in factor endowments and changes in the outputs of the two commodities when commodity prices are assumed to be given.

The factor content theorem makes the powerful prediction that free trade in goods actually equalizes the relative and absolute prices of homogeneous factors internationally. The essential reason for this is that trade in goods can substitute for trade in factors. There are numerous reasons why we do hot observe such equalization in practice, but there are important tendencies in that direction to the extent that international trade is the result of variations in factor endowments.

The factor-price-equalization theorem also generalizes for any number of factors and goods as long as the number of goods is equal to the number of factors and all countries produce all these goods (that is, they are incompletely specialized").

Q. Briefly discuss Stolper Samuelson theory

The Stolper–Samuelson theorem is a basic theorem in Heckscher–Ohlin trade theory. It describes the relationship between relative prices of output and relative factor rewards specifically, real wages and real returns to capital.

The theorem states that under specific economic assumptions (constant returns to scale, perfect competition, equality of the number of factors to the number of products) a rise in the relative price of a good will lead to a rise in the return to that factor which is used most intensively in the production of the good, and conversely, to a fall in the return to the other factor.

The Stolper–Samuelson theorem is closely linked to the factor price equalization theorem, which states that, regardless of international factor mobility, factor prices will tend to equalize across countries that do not differ in technology. It is important to examine this result further. The Stolper-Samuelson model holds factor endowments fixed while allowing commodity prices to change, which shifts the demands for factors, thereby changing real factor incomes. This is not overly restrictive, for it seems reasonable to suppose that factor supplies arc exogenously given. However, as we suggested earlier, there may be some question about this if, for example, laborers are able to clause between work and leisure.

Q. Briefly explain the term ‗gain from trade‟.

A sufficient condition for gains from trade is that consumption of the tax-distorted good increases. To put the matter the other way around, a country may fail to gain from trade when trade further reduces the consumption of a taxed good (or increases consumption of a subsidized good). Another popular view is that if one country gains through trade, the other country must lose. This is what economists would call a —zero sum game‖: the gains to one player equal the losses to the other player. There is a wide range of circumstances in which all countries gain mutually from trade, circumstances in which trade is a —positive-sum game‖.

Q. Identify and discuss the assumptions of the Heckscher – Ohlin theory of factor endowment.

Given the assumption of the model, a country will export the commodity that intensively uses its relatively abundant factor.

Note carefully the implication of this theorem. The important characteristics distinguishing each country the its relative supplies of capital and labor, By virtue of exporting the capital-intensive good and importing the labor-intensive good, Country H implicitly exports the services of capital, its abundant factor and imports the services of labour its scarce factor. Thus, international trade in commodities accomplishes the task of exchanging surplus factor services between countries. This is an important phenomenon for understanding the effects of trade on factor incomes.

An interesting question regarding the Heckscher-Ohlin theorem relates to how far we can relax the underlying assumption find still ensure that the result holds. Non-constant returns to scale would clearly invalid air the notion that country size is irrelevant for trade patterns, and allowing arbitrary international differences in technology would also render the theorem generally invalid. If factors were not homogeneous, meaning that labor was distinguished by skills and that capital came in different types, the simple two-factor theorem would no longer be relevant. However, as we discuss in point 8.6, an important variant of the fundamental message of the model that countries export the services of their abundant factors is still valid. If factors are not mobile between industries but instead must remain lived in employment for some period of time, the model must depart from the long-run nature of the Heckscher-Ohlin theorem in favor of a short-run view.

The existence of some market distortions, such as monopoly and export subsidies, can overturn the Heckscher-Ohlin theorem if their effects are powerful enough to offset the influence of endowments. However, the most common distortions we consider in trade models are import restrictions

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such as tariffs, quotas, and transport costs, which we analyze in later chapters. These barriers can reduce or even eliminate trade in goods, but any remaining trade would obey the pattern generated by the Heckscher-Ohlin model. Thus, the theorem is still applicable when trade restrictions are allowed.

Q. Clearly explain Production Possibility Frontier

The production possibility frontier (PPF) represents the point at which an economy is most efficiently producing its goods and services and, therefore, allocating its resources in the best way possible. If the economy is not producing the quantities indicated by the PPF, resources are being managed inefficiently and the production of society will dwindle. The production possibility frontier shows there are limits to production, so an economy, to achieve efficiency, must decide what combination of goods and services can be produced. The production possibility frontier, as its name implies, is a locus that shows all possible efficient production points. It is important to note that two kinds of efficiency are being assumed here. The first, which we might call engineering efficiency, implies simply that for either of the production functions and for any bundle of inputs, output is as large as it could possibly be.

Q. What do you understand by the pattern of trade?

The pattern of trade in the specific-factor model, each country will export the good with the absolutely abundant stock of specific capital, assuming identical endowments of labor, the mobile factor. With differences in labor endowments, trade patterns will depend on the nature of the production functions and on the allocation of capital (that is, on the stocks of specific factors. An understanding of trade requires an understanding of complex and indirect relationships, such as how a country's endowments of capital and labor determine its optimal pattern of trade. Before we can grasp the whole picture, we need to establish an understanding of the individual pieces of the puzzle. The ideas developed here will he used repeatedly throughout, the course material, so your investment should pay off. Its practical importance is in showing that, under reasonably general circumstances, differences in relative endowments still determine comparative advantage. However, comparative advantage in this sense refers to the pattern of trade in factor services rather than in goods.

Q. Explain briefly the concept of factor endowment

We define factor endowments specifically in terms of the ratios between capital stocks and labor forces in the two countries. Thus, if the capital-

labor ratio in Country H is greater than it is in Country F, we say that Country H is relatively capital abundant (and labor-scarce) while Country F is relatively labor-abundant (and capital scarce).

To understand the concept of relative factor endowments, consider the estimates of real capital endowments and labor forces. Capital stocks were computed as the cumulative sum of gross fixed capita! formation in the 15-year period through 1984, corrected for depreciation and inflation, and converted to U.S. dollars using a consistent set of international price and exchange rate comparisons.'' Thus, capital stocks are in billions of 1984 dollars. The labor force in each country is defined to be the economically active population (that is, those employed and those looking for work)

Q. Write short note on the following:

a. Factor price equalization

The factor-price-equalization theorem stated that in free trade with equalized commodity prices, if two countries produce both goods in common, real factor prices will become identical as well. The existence of different factor endowments does not change this result as long as countries are incompletely specialized. The theory relies on the assumption that there are at least as many goods as factors so that equalized goods prices would be sufficient to determine equalized factor prices. In this context it is no surprise that factor price equalization, ice; lot hold in the specific-factors model, because it embodies three factors and only two goods. To make the point more concede simply imagine countries H and F moving from autarky to free trade. If exports good X, then the real income of capital will rise and that of S-capital will fall, as we demonstrated. In F the real income of-S'- capital will rise and that of R-capital will fall as that country exports good Y, Labor will have a higher real wage with respect to good Y in H but a lower real wage with respect to good F in F (and the opposite effects would manifest with respect to good X}. There is clearly no presumption toward equalization of real factor prices in the short run.

b. Factor market distortion

The particular distortion we choose is motivated by the corpora's e income tax. The effects of this tax are significantly different from those of a commodity tax, for the corporate income tax applies to factor returns, which introduces a distortion into the production side of the model as well as changing relative commodity prices. In the following discussion we assume a two-good, two-factor, Heckaeher-Ohlin-type model, and we assume that the

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tax is imposed on one of the factors (capital) in one of the two industries (the corporate sector).

Three main effects of the corporate income tax are of interest. First, the corporate income tax, by creating a divergence between the factor-price ratios faced by producers in the two sectors, will result in production inside the production possibilities frontier. Second, the new distorted production "frontier" no longer must be concave to the origin but could he convex or could alternate between being concave and convex. Third, equilibrium price lines will generally not be tangent to the distorted production frontier.

c. Gains from exchange

The gains from trade can be conveniently broken down into gains from two distinct sources: gains from exchange and gains from specialization. The gains from exchange refer to the fact that if individuals or countries are endowed with different amounts of goods or have different preferences, they can both gain by trading with each other. For example, Janet could give Jim three bags of peanuts for two beers. They would then move to point C in their respective diagrams, each attaining a utility level of Uf . An implicit trading price is established, in so far as three bags of peanuts are deemed to be equal in value to two beers. Beer is relatively more valuable; the price of beer in terms of peanuts is 3/2 = 1.5.

Q. What is International Economics?

International economics can be divided conveniently into two parts: real analysis or trade theory, and monetary analysis or international finance. Real analysis studies the reasons that trade takes place, the implications for commodity and factor price of changes in real variables (such as the stock of capital and the supply of labor), the benefits that accrue from international trade, and the effect of trade restrictions on the welfare of the economy. Because its focus is equilibrium determination of real trade flows and welfare, trade theory generally analyzes barter exchange expressed in terms of a numeraire good. It ignores macroeconomic disequilibrium problems by assuming the existence of full employment and aggregate trade balance. Monetary analysis, on the other hand, is concerned with such issues as the determination of exchange rates and the international transmission of unemployment and inflation.

Often the two branches of international economics use different methodologies, with trade theory using market-clearing microeconomic equilibrium processes and international finance using macroeconomic concepts such as a single aggregate output and price level, in which there

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can be shortrun fluctuations. However, this distinction can easily he overdrawn. In recent years economists have made great strides in integrating the two approaches by modelling aspects of international finance, such as the existence of an aggregate trade deficit, as the result of microeconomic equilibrium processes in which agents trade goods both across borders and over time.

Q. Considering wage and the Ricardian framework, discus in detail the ‗wage -fund theory.'

Smith said that the demand for labour could not increase except in proportion to the increase of the funds destined for the payment of wages. Ricardo maintained that an increase in capital would result in an increase in the demand for labour. Statements such as these foreshadowed the wages-fund theory, which held that a predetermined ―fund‖ of wealth existed for the payment of wages. Smith defined this theoretical fund as the surplus or disposable income that could be used by the wealthy to employ others. Ricardo thought of it in terms of the capital—such as food, clothing, tools, raw materials, or machinery—needed for conditions of employment. The size of the fund could fluctuate over periods of time, but at any given moment the amount was fixed, and the average wage could be determined simply by dividing the value of this fund by the number of workers.

Regardless of the makeup of the fund, the obvious conclusion was that when the fund was large in relation to the number of workers, wages would be high. When it was relatively small, wages would be low. If population increased too rapidly in relation to food and other necessities (as outlined by Malthus), wages would be driven to the subsistence level. Therefore, went the speculation, labourers would be at an advantage if they contributed to the accumulation of capital to enlarge the fund; if they made exorbitant demands on employers or formed labour organizations that diminished capital, they would be reducing the size of the fund, thereby forcing wages down. It followed that legislation designed to raise wages would not be successful, for, with only a fixed fund to draw upon, higher wages for some workers could be won only at the expense of other workers.

This theory was generally accepted for 50 years by economists such as Nassau William Senior and John Stuart Mill. After 1865 the wages-fund theory was discredited by W.T. Thornton, F.D. Longe, and Francis A. Walker, all of whom argued that the demand for labour was not determined by a fund but by the consumer demand for products. Furthermore, the proponents of the wages-fund doctrine had been unable to prove the existence of any kind of fund that maintained a predetermined relationship with capital, and they also failed to identify what portion of the labour

force‘s contribution to a product was actually paid out in wages. Indeed, the total amount paid in wages depended upon a number of factors, including the bargaining power of labourers. Despite these telling criticisms, however, the wages-fund theory remained influential until the end of the 19th century.

Q. What is a trading economy, and when is it at equilibrium?

A trading economy is able to sell some of one good at world prices and use the proceeds to buy the other commodity. Instead of market clearing, we have what we call a trade balance condition: the value of what a country sells on world markets must be equal to what it buys. We can define the excess demand for good X and Y as (Xc – Xp) and (Yc – Yp) respectively. If excess demand is positive, the economy is consuming more than it is producing, which corresponds to demand for an import good. If excess demand is negative, an economy is consuming less than it is producing, resulting in an export good. The trade balance constraint requires that the value of all imports be equal to the value of all exports. An alternative way of saying this is that the sum of the value of the country‘s excess demands must equal zero: the positive excess demand for the import good must equal the negative excess demand for the export good.

We can think of the value of production as the income of the country. By placing a line with the slope of the world price ratio p* through the production point, we derive the —national budget line||. As we have discussed earlier, this budget line defines national income by evaluating domestic output at world prices. Consumers are then free to choose any point on this budget line, because the value of consumption will be equal to the value of production. This is shown in Fig. 4.2, where the fixed world price ratio is given by p*. Producers optimize by choosing production at point Q. Consumers optimize by choosing consumption at point C. In the particular case shown, the country imports X(Xc > Xp) and exports Y(Yc < Yp). Trade will balance insofar as the value of production at world price equals the value of consumption. To summarize, the conditions for general equilibrium are given as follows.

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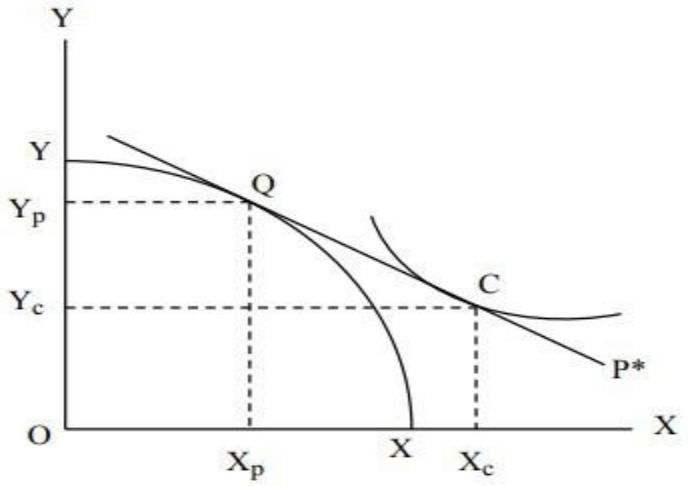



Figure 4.2: Open-economy general equilibrium

Note finally that the autarky market clearing condition is a special case of trade balance. It satisfies the trade balance condition in that both terms in parentheses are zero. If world prices happened to be the same as the country's autarky prices, then the trading equilibrium would be identical to the autarky equilibrium.

Q What is another name for Heckscher – Ohlin – Vanek theorem?

Heckscher-Ohlin-based exchange is often termed inter-industry trade

Q. Differentiate between real analysis and monetary analysis

Real analysis studies the reasons that trade takes place, the implications for commodity and factor price of changes in real variables (such as the stock of capital and the supply of labor), the benefits that accrue from international trade, and the effect of trade restrictions on the welfare of the economy. Because its focus is equilibrium determination of real trade flows and welfare, trade theory generally analyzes barter exchange expressed in terms of a numeraire good. It ignores macroeconomic disequilibrium problems by assuming the existence of full employment and aggregate trade balance.

Monetary analysis, on the other hand, is concerned with such issues as the determination of exchange rates and the international transmission of unemployment and inflation. Often the two branches of international economics use different methodologies, with trade theory using market-clearing microeconomic equilibrium processes and international finance using macroeconomic concepts such as a single aggregate output and price level, in which there can be shortrun fluctuations. However, this distinction can easily he overdrawn. In recent years economists have made great strides in integrating the two approaches by modelling aspects of international

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finance, such as the existence of an aggregate trade deficit, as the result of microeconomic equilibrium processes in which agents trade goods both across borders and over time.

Q. Explain the concept of _Constant Returns to Scale'

A very common assumption in economics is that of constant returns to scale, the assumption that proportional changes in all inputs lead to the same proportional change in output. This assumption is referred to somewhat more formally as homogeneity of the first degree. It is such an important concept in economics and in the discussion of this material that a formal definition seems worthwhile. Let $\lambda > 0$. The function $X = F(K, L)$ is said to be homogeneous of degree k if $\lambda X = F(\lambda K, \lambda L)$. If $k = 1$, the function is said to be homogeneous of degree 1, and production is characterized by constant returns to scale.

Q. Define the term _specific factor'

A specific factor is one which is stuck in an industry or is immobile between industries in response to changes in market conditions. A factor may be immobile between industries for a number of reasons. Some factors may be specifically designed (in the case of capital) or specifically trained (in the case of labor) for use in a particular production process. The specific factor model is designed to demonstrate the effects of trade in an economy in which one factor of production is specific to an industry. The most interesting results pertain to the changes in the distribution of income that would arise as a country moves to free trade.

Q. What is general equilibrium theory?

In economics, general equilibrium theory attempts to explain the behavior of supply, demand, and prices in a whole economy with several or many interacting markets, by seeking to prove that the interaction of demand and supply will result in an overall general equilibrium. General equilibrium theory contrasts to the theory of partial equilibrium, which only analyzes single markets.

General equilibrium theory studies economies using the model of equilibrium pricing and seeks to determine in which circumstances the assumptions of general equilibrium will hold. The theory dates to the 1870s, particularly the work of French economist Léon Walras in his pioneering 1874 work Elements of Pure Economics. General equilibrium theory tried to show how and why all free markets tended toward equilibrium in the long run. General equilibrium theory builds on the coordinating processes of a

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free market price system, first widely popularized by Adam Smith's —The Wealth of Nations|| (1776).

Broadly speaking, general equilibrium tries to give an understanding of the whole economy using a "bottom-up" approach, starting with individual markets and agents. (Macroeconomics, as developed by the Keynesian economists, focused on a "top-down" approach, where the analysis starts with larger aggregates, the "big picture".) Therefore, general equilibrium theory has traditionally been classified as part of microeconomics.

Q. What are the conditions that determine general equilibrium in a closed economy?

Of the three conditions that determine general equilibrium in a closed economy, the first two are optimization conditions for producers and consumers:

(1) Competitive, profit-maximizing producers pick outputs such that, at given commodity prices, the marginal rate of transformation is equal to the producer price ratio; this condition was given in Eqs. (2.13) as Px/Py = MRT.

(2) Consumer pick commodities such that, at given commodity prices, their marginal rate of substitution 'in consumption is equal to the consumer price ratio; assuming that consumer and producer prices are the same, this condition was given in Eq. (3.4) as Px/Py = MRS. The third condition is a market clearing condition:

(3) The supply and demand for each commodity must be equal; let subscript c denote consumption of a commodity and subscript p denote production of a commodity.

Our three conditions for general equilibrium are summarized by:

Px/Py = MRT Producer optimization

Px/Py = MRS Consumer optimization

Xc = Xp Yc = Yp Market clearing

Q. List the two divisions of gains from trade

The gains from trade can be conveniently broken down into gains from two distinct sources:

1. Gains from exchange
2. Gains from specialization

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Q. Define gains from exchange

The gains from exchange refer to the fact that if individuals or countries are endowed with different amounts of goods or have different preferences; they can both gain by trading with each other.

Q. Mention the three conditions that determine the position and shape of the production possibility curve

On the production side three conditions determine the position and shape of the production possibility curve:

1. Degree of homogeneity
2. Factor endowments
3. Production functions

Q. Give three examples of distortions in the no-trade model

1. Distortion include
2. Taxes
3. Subsidies
4. Imperfect competition

Q. List the set of conditions that simultaneously guarantee the no-trade situation

1. Identical production functions among countries

2. The same relative endowments in all countries

3. Constant returns to scale

4. Identical and homogenous taste in all countries

5. The absence of distortions (e.g., taxes, subsidies, imperfect competition)

Q. List the effects of international trade

1. Dual economies

2. Not much beneficial for poor countries

3. Limited possibility of gain

4. Adverse effect on _demonstration effect‘

5. Secular deterioration in the terms of trade

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Q. Explain the consequence of adverse effect on demonstration effect of international trade

Another harmful effect is that the international operation of the ‗demonstration effect' has been a handicap for the poor countries. It has been responsible for reducing the capacity for capital formation. The desire for luxury, show-off for higher standard of living and patterns of consumption of advanced countries has been an important factor responsible for low level of domestic savings in underdeveloped countries. Higher income groups in these countries are trying to adopt the consumption standards of advanced countries which have pushed up their propensity to consume and thereby limited capital accumulation and economic growth. This leads to corruption and black marketing. Thus, these evils have adverse effect on the economy.

Q. Explain the subsistence theory of wages

Subsistence theories emphasize the supply aspects of the labour market while neglecting the demand aspects. They hold that change in the supply of workers is the basic force that drives real wages to the minimum required for subsistence (that is, for basic needs such as food and shelter). Elements of a subsistence theory appear in The Wealth of Nations, where Smith wrote that the wages paid to workers had to be enough to allow them to live and to support their families. The English classical economists who succeeded Smith, such as David Ricardo and Thomas Malthus, held a more pessimistic outlook. Ricardo wrote that the ―natural price‖ of labour was simply the price necessary to enable the labourers to subsist and to perpetuate the race. Ricardo's statement was consistent with the Malthusian theory of population, which held that population adjusts to the means of supporting it.

Subsistence theorists argued that the market price of labour would not vary from the natural price for long: if wages rose above subsistence, the number of workers would increase and bring the wage rates down; if wages fell below subsistence, the number of workers would decrease and push the wage rates up. At the time that these economists wrote, most workers were actually living near the subsistence level, and population appeared to be trying to outrun the means of subsistence. Thus, the subsistence theory seemed to fit the facts. Although Ricardo said that the natural price of labour was not fixed (it could change if population levels moderated in relation to the food supply and other items necessary to maintain labour), later writers were more pessimistic about the prospects for wage earners. Their inflexible conclusion that wages would always be driven down earned the subsistence theory the name ―iron law of wages.‖

Q. Elucidate the wages-fund theory

Smith said that the demand for labour could not increase except in proportion to the increase of the funds destined for the payment of wages. Ricardo maintained that an increase in capital would result in an increase in the demand for labour. Statements such as these foreshadowed the wages-fund theory, which held that a predetermined ―fund‖ of wealth existed for the payment of wages. Smith defined this theoretical fund as the surplus or disposable income that could be used by the wealthy to employ others. Ricardo thought of it in terms of the capital—such as food, clothing, tools, raw materials, or machinery—needed for conditions of employment. The size of the fund could fluctuate over periods of time, but at any given moment the amount was fixed, and the average wage could be determined simply by dividing the value of this fund by the number of workers.

Q. Explain the residual-claimant theory

The residual-claimant theory holds that, after all other factors of production have received compensation for their contribution to the process, the amount of capital left over will go to the remaining factor. Smith implied such a theory for wages, since he said that rent would be deducted first and profits next. In 1875 Walker worked out a residual theory of wages in which the shares of the landlord, capital owner, and entrepreneur were determined independently and subtracted, thus leaving the remainder for labour in the form of wages. It should be noted, however, that any of the factors of production may be selected as the residual claimant—assuming that independent determinations may be made for the shares of the other factors. It is doubtful, therefore, that such a theory has much value as an explanation of wage phenomena.

Q. Explain the bargaining theory

The bargaining theory of wages holds that wages, hours, and working conditions are determined by the relative bargaining strength of the parties to the agreement. Smith hinted at such a theory when he noted that employers had greater bargaining strength than employees. Employers were in a better position to unify their opposition to employee demands, and employers were also able to withstand the loss of income for a longer period than could the employees. This idea was developed to a considerable extent by John Davidson, who proposed in The Bargain Theory of Wages (1898) that the determination of wages is an extremely complicated process involving numerous influences that interact to establish the relative bargaining strength of the parties.

This theory argues that no one factor or single combination of factors determines wages and that no one rate of pay necessarily prevails. Instead, there is a range of rates, any of which may exist simultaneously. The upper limit of the range represents the rate beyond which the employer refuses to hire certain workers. This rate can be influenced by many factors, including the productivity of the workers, the competitive situation, the size of the investment, and the employer‘s estimate of future business conditions. The lower limit of the range defines the rate below which the workers will not offer their services to the employer. Influences on this rate include minimum wage legislation, the workers‘ standard of living, their appraisal of the employment situation, and their knowledge of rates paid to others. Neither the upper nor the lower limit is fixed, and either may move upward or downward. The rate or rates within the range are determined by relative bargaining power.

Q. Briefly discuss the role of wages in the Ricardian Framework

Ricardian model, comparative advantage is determined simply by the relative productivity of labor in producing commodities, or, equivalently, by international differences in production functions. It may seem surprising wage rates did not enter the discussion. After all there has been much concern expressed in high-income economies about the possible effects of competition from low-wage workers in developing countries. In his unit we examine the role of wages in the Ricardian framework.

$P\ a\ x\ \alpha h = wh\ Pa\ y\ \beta h = wh$ (7.8)

Here, a and β are the marginal products of labor in goods X and Y. It immediately follows that the relative price in autarky is independent of the wage fate:

$P\ a\ h = \beta h/\alpha h$ (7.9)

This equation reflects what Ricardo referred to as the Labor Theory of Value. Relative prices must equal relative real costs in terms of labor inputs. Here, if sector Y has a relatively low marginal product and sector X has a relatively high marginal product, the home country is likely to have its comparative advantage in good X. The wage rate, which is the nominal price of labor, has no effect on relative commodity prices as long as it is the same in both sectors.

The wage rate is relevant for determining real wages, or the living standards of laborers. Note from Kqs. (7.8) that real wage rates equal marginal labor productivities:

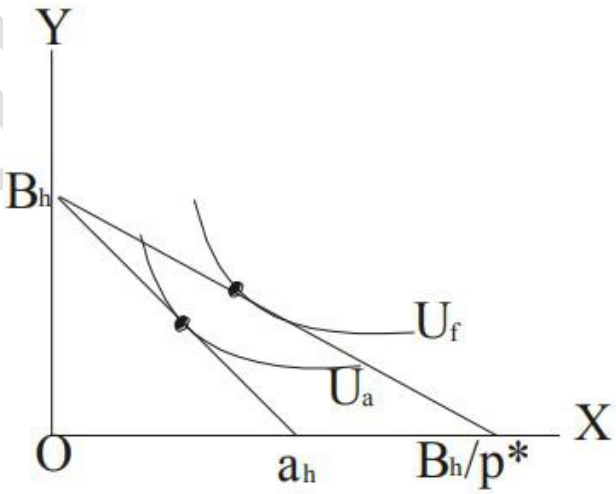
wh / pa x = ah wh/pa y = βh (7.10)

These "real wages" can be interpreted graphically as the end points on the budget line of an individual worker. Assume that 1 worker owns 1 unit of labor. & is the maximum amount of Y that can be purchased if all income is spent on Y; similarly, a is the maximum amount of X that can be purchased if all income is spent on X. An individual worker's budget line is given in Fig. 7.5.

Consider again the movement from autarky to free trade. This change will alter nominal wages in both countries, generating wage rates Wh and Wf. To determine the impacts on rear wages, note that in free trade the home country exports good Y and the foreign country exports good X. Since the free trade prices of each good must equal the average cost of producing them, we know that real wages for a home laborer are constant in terms of good Y. However, the worker can now purchase good X at price ratio p < βh/αh. Because the price ratio p" is the vector intercept of the budget line (β) over the horizontal distance, the horizontal intercept (i.e., the wage wh/P* X) must be βh/p* > αh. The intercepts of the consumer's line free trade are now given by

wh / pa x = βh/p* > αh wh/p* y = βh (7.11)

Thus, trade does not alter the home country's real wage in terms of its export good, but it does change the real wage in terms of its import good. The free trade budget line is shown in Fig, 7.5, where we see that the welfare of an individual worker rises from Uα„ to Uf . A corresponding argument can be made about the welfare of an individual worker in Country F' whose income is fixed in terms of good X but rises in terms of Y .



There is an important final point to make. It is quite possible that in free trade, real wages are higher in one country than in the other. We can show that if the home country, for example, has an absolute advantage in both goods, it will necessarily have a higher real wage than does the foreign country. In such a situation, we have αh> αf , where αf is a worker in Country F's real wage in terms of X. Using αh > αf together with Eqs. (7.11), we have

wh /P* x > αh > αf = wf /P*x (7.12)

We can similarly show that a worker in Country H earns a higher wage in terms of good Y when H has this absolute advantage in both goods. In summary, absolute advantage is important for determining diligences in real wages (per capita incomes) across countries, but comparative advantage determines the direction of trade.

Q. Define trade balance

Balance of trade (BOT) is the difference between the value of a country's imports and exports for a given period and is the largest component of a country's balance of payments (BOP). Economists use the BOT to measure the relative strength of a country's economy. The balance of trade is also referred to as the trade balance or the international trade balance. A country that imports more goods and services than it exports in terms of value has a trade deficit. Conversely, a country that exports more goods and services than it imports has a trade surplus. The formula for calculating the BOT can be simplified as the total value of imports minus the total value of exports.

Q. Discuss the analysis of two good and two factor model.

We assume that two commodities, X and Y, are produced using two factors, capital and labor, with technologies described by the production functions shown in Eqs. (2.8).

X = Fx, (Kx, Lx) (2.8)

Y = Fy, (Ky, Ly)

Note that subscripts are now being used to distinguish the two production functions and the inputs, used by each. These production functions are assumed to be homogeneous of the first degree and are assumed to be increasing functions of both inputs. It is further assumed that positive outputs imply positive inputs of both factors. The economy is assumed to have fixed total supplies of both capital and labor, and these two constraints are represented by Eqs. (2.9):

K = Kx + Kx (2.9)

L = Ly + Ly

As well as showing the allocation of the two factors between the two production processes, the equality sign in these two equations implies that, these two processes use all the available K and L. Full employment is, therefore, implicitly assumed. Also implicit in our analysis is the assumption that both factors of production are completely divisible and are homogeneous in the sense that the units are identical.

An assumption central to the analysis is that the commodities, X and Y, differ in the sense that the production function differ. Representative isoquants Y0 and X0 for the two industries are shown in Fig. 2.6. While it is clear from the diagram that these two isoquants have been derived from different production functions, it will be useful to describe the differences in a somewhat more formal manner. Consider an arbitrary wage-rental ratio equal to the slope of the line K0L0. With these relative factor prices, production in industry Y would take place somewhere along the line OA, and production in industry X somewhere along the line OB, these being the points where the wage-rental ratios are tangent to the respective isoquants.

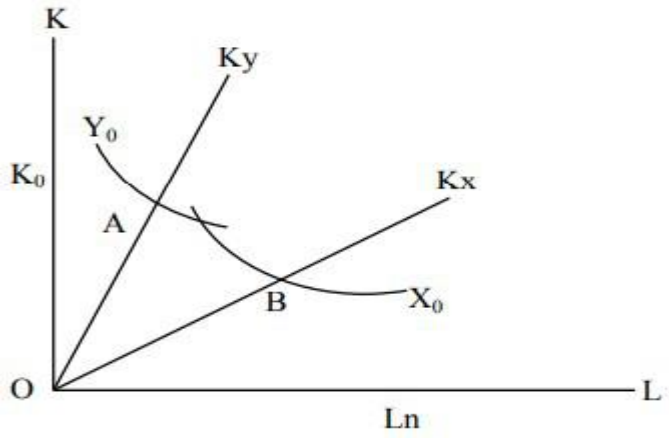



Fig. 2.6: Factor intensities

It is clear that the capital-labor ratios k = K/L for industries Y and X, represented in the figure by Ky and Kx respectively, differ. We thus have the following definition.

Definition. Consider fixed factor prices. If Ky > Kx, at those factor prices, Y is said to be capital intensive and X is said to be labor intensive.

From Fig. 2.6 it is evident that Ky is greater than Kx, and thus commodity Y is said to be capital-intensive relative to commodity X as noted in the

definition. Of course, a completely equivalent statement is that commodity X is labor-intensive relative to commodity Y. For the remainder of the book, we assume that commodity Y is capitalintensive relative to commodity X for all wage-rental ratios. This is known as the strong factor intensity hypothesis.

Q. Discuss the effect of Government policy from gain from trade analysis.

Gains from trade, as a result of Government Policy which means government policy has a lot to do with the benefit of trade among countries. However, it can be described as a result of specialization in production from division of labor, economies of scale, scope, and agglomeration and relative availability of factor resources in types of output by farms, businesses, location and economies, a resulting increase in total output possibilities, trade through markets from sale of one type of output for other, more highly valued goods. Trade induced by the introduction of a distortion such as a commodity tax or subsidy is welfare-reducing trade. Governments must be careful not to confuse exports with welfare. Government policies can induce the export of a good, but such exports are welfare-reducing in an initially competitive, distortion-free economy. However, the consumption and production taxes or subsidies are quite different from each other in an open economy. In particular, they have opposite effects on the direction of trade. A consumption tax decreases the consumption of a good, leading to increased exports or decreased imports. A production tax decreases the production of the good, leading to decreased exports or increased imports.

Q. Discuss the analysis of Consumer, Producer and World Prices

When we introduce taxes and subsidies into the analysis, it becomes important to distinguish prices gain by consumers from prices received by producers. Once we introduce trade, consumer and producer prices must be distinguished from world prices, the prices at which the country can trade. Throughout this chapter, we will use the notation q to represent consumer prices, p to represent producer prices, and p' to represent world prices. This notation will also be used in later chapters, particularly the chapter on tariffs.

In order to focus on trade issues, we will also make the assumption throughout this unit that there is no government sector per se; the government returns all tax collections to consumers in lump-sum fashion and/or raises all subsidies by lump-sum taxation. We implicitly assume a very large number of consumers, with each consumer getting a check or a bill which give to (or takes from) the consumer his or her share of taxes subsidies). Consigners regard their bub or checks as being unaffected by their own purchases. For example, if a consumer pays $1 in sales tax, the

consumer gets a refund of only $1/N of that amount where .Vis the number of consumers (consider the refund if there are 100 million consumers). Thus each consumer does indeed regard the tax as raising prices, even though the tax is returned to all consumers collectively. Similar comments apply to subsidies.

Throughout the chapter we will specify taxes and subsidies in an ad valorem (percentage of value) form rather than in specific form, t will denote a tax and a subsidy. Ad valorem taxes are quoted as rates. A sales tax of 5 percent, for example, would mean a tax rate of; = .05 in this context. (Specific taxes, on the other hand, are quoted in monetary units per unit of the good: the US gasoline tax is quoted in cents per gallon.).

Q. Discuss the relationship between Consumer and a Producer with a tax or subsidy.

Thus the relationship between consumer and producer prices with a tax or subsidy is given as follows.

$$q = p\,(1+t) > p \qquad \text{tax}$$
$$q = p\,(1-s) < p \qquad \text{subsidy} \qquad (10.1)$$

A tax raises the consumer price above the producer price, while a subsidy lowers the consumer price below the producer price. A tax rate t = .05 raises the consumer price 5 percent above the producer price: q = p (1.05). When there are only two goods, the effects of a tax on one good are equivalent to a subsidy on the other good. In order to see this, consider the commodity price ratios resulting from a tax on Y versus a subsidy to X.

$$\frac{q_x}{q_y} = \frac{P_x}{P_y(1+t)} < \frac{P_x}{P_y} \qquad \text{tax on Y}$$

$$\frac{q_x}{q_v} = \frac{P_x(1-s)}{P_v} < \frac{P_x}{P_v} \qquad \text{subsidy on X}$$

We see from Esq. (10.2) that a subsidy to X and a tax on Y induce the same "wedge" between the consumer and producer price ratios.

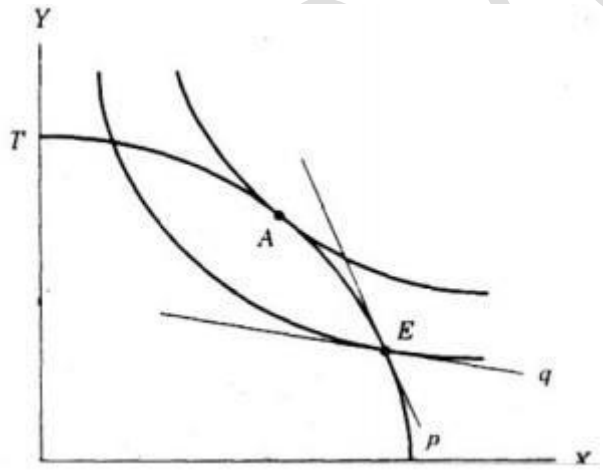
Figure 10.1 gives autarky equilibrium at point E when there is either a tax on Y or a subsidy un X, assuming that the tax revenue is redistributed in lump sum fashion and the subsidy is raised by a lump sum tax. These latter assumptions are reflected in the fact the consumption and production bundles are the same even though, in the case of a tax, for example, the consumption bundle costs more than the value of those goods at producer

prices. The consumers pay more than the producers receive because of the tax, but then they receive an income in excess of the value of production because they receive the tax refund. Lei subscripts c and p denote consumption and production quantities, respectively. For a tux on Y,

$$q_x X_c + q_y Y_c = p_x X_c + p_y(1 + t)Y_c = /p_x X_c + p_y Y_c/ + /P_y^{\mathrm{T}} Y_c/$$

The left-hand side of Eq. (10.31) is consumer expenditure at consumer price. The first bracketed term on the right-hand side is income received from production (payments of factors of production), while the second term on the right-hand side is redistributed tax revenue. Thus, consumer expenditure equals consumer income. A similar analysis of a subsidy requires only that we change the sign of t.

Figure 10.1 illustrates the distortionary effect of the Lax on Y or the subsidy on X. Welfare is lower at E than at the undistorted competitive equilibrium at A. The producer price ratio p is tangent to the production




frontier TT' , while the consumer price ratio q is tangent to the indifference curve through E. The tax causes the consumers to perceive Y as more expensive than its actual production cost, or a subsidy to X causes consumers to perceive as relatively cheaper than its actual production cost.

First, governments usually raise revenues in order to provide public goods (hat, are not or cannot be provided by markets. This analysis takes no account of public goods. Second, not all taxes are distortionary or as distortionary as the commodity tax shown here. For example, in the present model, an equal ad valorem tax on both goods would leave the relative consumer and producer prices equal. Such a set of taxes is non-distortionary. We will return to this point in discussing factor taxes later in the chapter.

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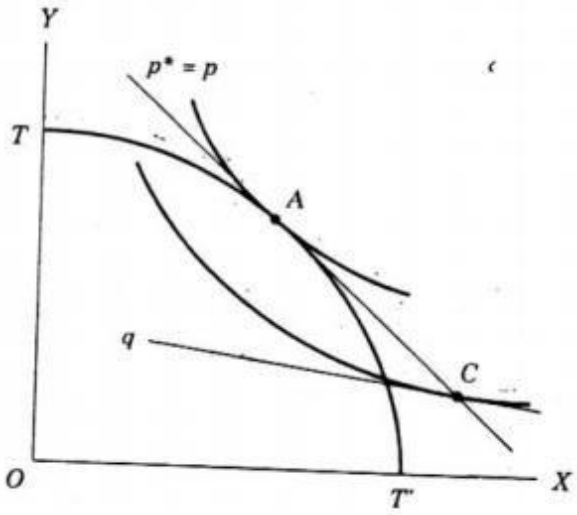
Finally, some government policies are imposed to correct an existing distortion in the economy, much as ;m environmental externality, in such a situation, Fig. 10.1 might accurately depict the effects of a pollution tax (on Yi on production and trade, hut the indifference curves no longer accurately indicate welfare changes. Welfare may be improving due to lower pollution (i.e., there is actually a third good, environmental quality, not shown in the diagram). More will be said about tuxes in the presence of existing.

Q. Discuss the analysis of taxes and subsidies as determinant of trade.

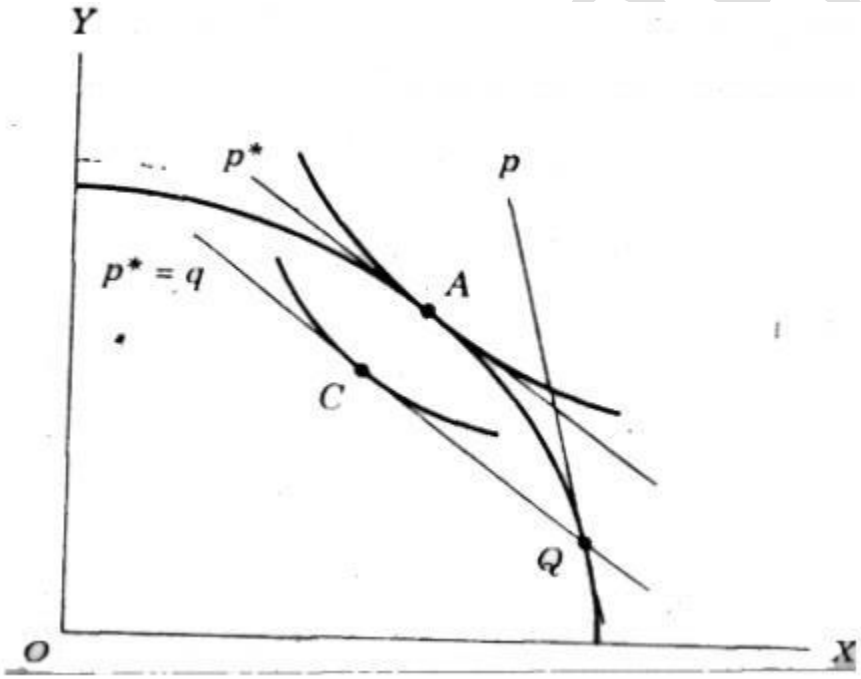
Suppose that Country II faces fixed world prices. Assume also that these prices happen to be equal to I 1's Autarky price ratio so that H does not choose to trade at these prices. The situation is shown in Figs. 10,2 and 10.3, where the autarky equilibrium A is also the free trade equilibrium at price ratio p.

Once we introduce trade, we have to keep track of not only consumer and producer prices, but also world prices. This in turn means that we have to specify whether a Lax or subsidy is assessed on consumption or production. In the closed economy it does not matter, because production and consumption of each good are equal. But with trade, consumption and production are generally not equal, so it matters which one we are taxing. With a consumption Lax, consumers pay a tax on both domestic and imported goods. Alternatively, if the good is exported, then the tax is paid only on the part of domestic production that stays in the country. Producers can trade at world prices. The relationships among consumer, producer and world prices with a consumption tax on Y are given by

$$\frac{q_x}{q_y} = \frac{P_x}{p_y(1+t} = \frac{P_x}{p_y(1+t)} \quad \text{or} \quad (1+t)\ \frac{q_x}{q_y} = \frac{p_x}{p_y} = \frac{p_x}{p_y}$$



The second equation emphasizes that producers in H face world prices. The consumption tax on Y (consumption subsidy on X) shown in Fig 10.2. When the tax is levied. Producers will continue to produce at A where the world price ratio, equal to the producer price ratio, is tangent to the production frontier. The balance-of-trade constraint requires trade to balance at world prices, so we know that the consumption point must be on p' through A. The consumption point is given by the point on that "national budget line" through A where the slope of an indifference curve is equal to the consumer price ratio. We show this in Fig, 10.2 as point C and label the consumer price ratio as q. Economic intuition would lead to the result shown for the consumption tax in Fig. 10.2 even without the formal analysis. We see that, given fixed




world prices, the tax has no effect on production but that consumption of) is discouraged. Consumers substitute away from the expens.ve commodity and in favor of the relatively cheaper commodity, X.

Welfare is reduced by this distortion, because consumers make efficient choice, when they do not face the true costs of producing cnmnnodit.es. But again, we should separate this welfare result from suits concerning consumption and trade, because not all consumption tax, must be welfare-reducing. Most countries have gasoline (petrol) example, which have the beneficial effects of reducing pollution and traffic congestion. Now considera tax on the production of Y or a subsidy on the production of X. In this case, consumers, not producers, face world prices. The relationships among the three price ratios are given by

$$\frac{P_x}{p_v(1+t)} \quad \text{or} \quad \frac{q_x}{q_v} = \frac{p_x}{p_v} \qquad (10.51$$

The relationship, in Eq. (10.5) are shown in Fig. 103. The Producer price ratio is now greater than the consumer and world price ratios, so p due on is show,, as taking place at point Q in Fig. 10.3. Consumption must take place along world price ratio through Q, and consumers now face world prices. Thus, the consumption point is given by the tangency between an indifference curve and the price line p* through point Q. we show the consumption point as C in fig. 10.3. The production tax discourage production of Y and leads to a substitution in production towards good X

The "second thing that Figs. 10.2 and 10.3 help to emphasize .s that production and consumption taxes/subsidies are very different from each other in the open economy. For example, they have opposite effect the direction of trade. A consumption tax on Y discourages consumption and therefore tends to lead to exports of Y (production minus consumption). A production tax on Y discourages production and therefore tend, lead to imports of Y (consumption on minus production). Governments must therefore be careful about where they levy a tax when assessing its likely effects.

Q. Write briefly on the following:

(i) Taxes

Taxes are involuntary fees levied on individuals or corporations and enforced by a government entity—whether local, regional or national—in order to finance government activities. In economics, taxes fall on whoever pays the burden of the tax, whether this is the entity being taxed, such as a business, or the end consumers of the business's goods. When the tax is levied producers will continue to produce at A where the world price ratio, equal to the producer price ratio, is tangent to the production frontier.

(ii) Subsidies

A subsidy or government incentive is a form of financial aid or support extended to an economic sector generally with the aim of promoting economic and social policy. Although commonly extended from government, the term subsidy can relate to any type of support – for example from NGOs or as implicit subsidies. The examination of taxes and subsidies is intended as an example of the effects that a wide range of government policies can have on trade. For example, environmental policies and regulations impact

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on firms' costs, therefore having effects on outputs and trade similar to those produced by taxes.

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